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RESEARCH ARTICLE

Enhancing Stock Market Prediction With Hybrid Deep Learning: Integrating LSTM, Transformer Attention, Federated Learning, and Sentiment Analysis

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ABSTRACT Accurate stock market prediction remains a critical yet challenging task due to the highly non-linear, volatile, and sentiment-driven nature of financial markets. In this paper, we present a hybrid deep learning framework that integrates long-short-term memory (LSTM) networks with Transformer-based attention mechanisms, sentiment analysis from financial news, and a privacy-preserving Federated Learning (FL) strategy. First, we benchmark traditional forecasting approaches, including ARIMA, SARIMAX, Prophet, Random Forest, and Support Vector Regression, against the baseline LSTM models. Our results show that LSTMs consistently outperform conventional methods in capturing temporal dependencies. To further enhance predictive accuracy, we incorporate Transformer attention to improve long-range dependency modeling and apply sentiment analysis using FinBERT-tone to embed market sentiment signals into the model. Finally, we simulate a Federated Learning environment, enabling decentralized model training without sharing raw financial data, thus addressing privacy concerns in the financial domain. Experimental results in ten major technology companies (Tesla, Apple, Amazon, Microsoft, Google, etc.) demonstrate that our hybrid model achieves superior short-term forecasting performance, with an average R^2 variance score of 0.91 across ten major technology companies and a trend precision of 65.36 %, demonstrating strong prediction performance for short-term stock forecasting. These findings highlight the potential of combining deep sequential models, attention mechanisms, and privacy-sensitive training strategies for robust and secure stock market forecasting.

INDEX TERMS Stock market prediction, long-short-term memory (LSTM), transformer attention, federated learning, sentiment analysis, financial time series forecasting, deep learning in finance, privacy preserving.

I. INTRODUCTION

The stock market, as a cornerstone of the global economy, exhibits extreme volatility, nonlinearity, and sensitivity to external events such as news sentiment and investor behavior. Predicting market fluctuations has long been a great challenge for researchers and practitioners due to the intricate interplay of temporal dependencies, external signals, and high-frequency variations. Accurate forecasts can provide

significant benefits for investors, financial institutions, and policy makers, yet conventional approaches -ranging from statistical models such as ARIMA and SARIMAX [1] to machine learning methods such as Random Forest [2] and Support Vector Regression [3] often struggle to capture the complex temporal dynamics and sentiment-driven patterns inherent in financial data.

Recent advances in deep learning, particularly recurrent architectures such as long-short-term memory (LSTM) networks [4], [5], have shown substantial promise in modeling sequential dependencies for time-series forecasting. LSTMs

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are particularly well-suited for financial prediction tasks, as they mitigate the vanishing gradient problem in traditional recurrent networks and preserve long-term contextual information. However, several challenges remain. First, LSTM models often fail to fully capture long-range dependencies across time steps, which may reduce performance in highly dynamic markets. Second, financial markets are heavily influenced by investor sentiment and news signals that are not adequately represented by historical price data alone [6], [7]. Finally, model training in finance raises growing privacy concerns, as traditional centralized data aggregation may expose sensitive institutional or investor-level information [8], [9].

A. RELATED WORK

A substantial body of research has investigated the application of machine learning to financial forecasting. Traditional statistical models, including ARIMA and SARIMAX, remain common baselines [1]. More recent approaches, on the other hand, employ Support Vector Regression [3], Random Forest [2], and gradient boosting methods.

Deep learning has further advanced the field: LSTM-based approaches [4], [5] have achieved state-of-the-art performance in capturing sequential dependencies in stock prices. For example, Li et al. introduced DP-LSTM [10], which integrates differential privacy principles into stock market forecasting using financial news, demonstrating improved accuracy while preserving data confidentiality. Similarly, transformer-based architectures have been applied to predict stock trends [11], [12], using attention mechanisms to capture long-range dependencies in financial time series. Recent studies continue to extend this line of work. Yang et al. [13] provide a systematic evaluation of deep learning models for stock prediction, while Li et al. [14] propose an explainable deep learning framework to enhance interpretability in financial forecasting.

Parallel research has highlighted the importance of sentiment analysis from financial news and social media, demonstrating that textual signals can significantly improve forecasting when combined with quantitative data. Similarly, privacy-preserving approaches, such as differential privacy and federated learning, underscore the need for scalable frameworks that facilitate collaboration without requiring the sharing of raw data. Despite these advances, few studies have systematically integrated LSTM, Transformer attention, sentiment analysis, and federated learning into a unified framework for stock market prediction. To address this gap, our work adopts both numerical metrics (R^2 , MAE, MSE) and directional accuracy (Mean Prediction Accuracy, MPA) to provide a more comprehensive evaluation.

Stock market dynamics are driven not only by nonlinear temporal patterns in historical prices, but also by exogenous factors such as news and investor sentiment, while practical deployment increasingly requires privacy-aware learning across distributed data sources. To address these

challenges in a unified framework, this work integrates four complementary components. First, long short-term memory (LSTM) networks are employed to capture sequential dependencies in historical OHLCV (Open, High, Low, Close, and Volume) data. Second, multi-head Transformer attention is incorporated to enhance long-range dependency modeling and selectively emphasize informative time steps within the price sequence. Third, FinBERT-based sentiment features are used to inject market-relevant textual signals derived from financial news, which are not directly observable from price history alone. Finally, a federated learning strategy enables privacy-preserving, decentralized model training across heterogeneous data silos while improving robustness and generalization. By jointly leveraging temporal modeling, attention mechanisms, sentiment awareness, and distributed learning, the proposed design overcomes key limitations of purely price-based and centralized stock forecasting models.

B. CONTRIBUTIONS

To address these gaps, this article makes the following key contributions.

- Comprehensive comparison of traditional statistical and machine learning models (ARIMA, SARIMAX, Prophet [15], Random Forest, SVR) against baseline LSTM networks for stock prediction.
- Hybrid architecture that integrates LSTM with multi-head transformer attention [11], enhancing the modeling of long-range temporal dependencies.
- Sentiment-aware prediction using FinBERT tone [6] to capture qualitative signals from financial news headlines.
- Privacy-preserving training through a simulated federated learning environment [8], [9], which allows decentralized learning without exposing raw financial data.
- Extensive validation on ten major technology companies (Tesla, Apple, Amazon, Microsoft, Google, etc.), demonstrating superior short-term forecasting with an average R^2 of 0.91 and a trend prediction accuracy of 65.36%.

By combining sequential modeling, attention mechanisms, sentiment analysis, and privacy-aware training, this work advances the development of robust, interpretable, and secure financial forecasting systems.

C. ORGANIZATION

The remainder of this paper is organized as follows. Section II provides background on long-short-term memories (LSTM) networks, sentiment analysis, and federated learning. Section III introduces the proposed framework, including the dataset, the architecture of the sentiment-enhanced LSTM model, and the integration of federated learning for privacy-preserving training. Section IV describes the experimental setup, evaluation metrics and comparative analysis against traditional forecasting models and a baseline

LSTM, followed by an exploration of the impact of Transformer attention, sentiment analysis, and federated learning on model complexity and predictive performance. Finally, Section V concludes the article, summarizing key findings, discussing implications for financial forecasting, and presenting directions for future research.

II. BACKGROUND

A. LSTM-BASED APPROACH

Long-short-term memory (LSTM) networks are a specialized type of recurrent neural network (RNN) introduced by Hochreiter and Schmidhuber [4] and later extended in surveys such as [5]. They were designed to address the limitations of standard RNNs in modeling long-range dependencies in sequential data, such as stock prices. Unlike traditional RNNs, which often suffer from vanishing or exploding gradients, LSTMs incorporate a gating mechanism that regulates the flow of information, enabling them to selectively retain or discard features over time steps. This property makes LSTMs particularly well-suited for financial time series forecasting [16], [17].

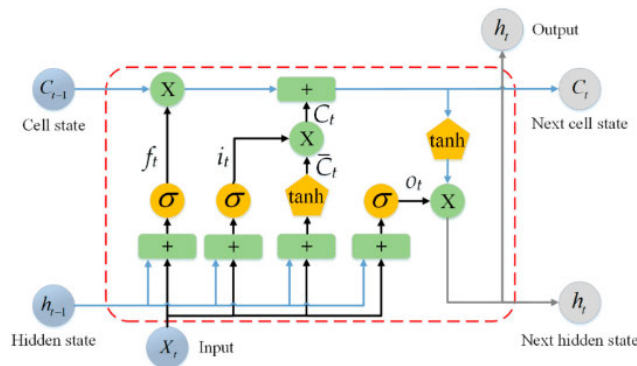


FIGURE 1. Long Short-Term Memory (LSTM) cell structure, including the forget gate, input gate, output gate, and cell state [4], [5].

As illustrated in Fig. 1, an LSTM cell receives two inputs at each time step: the current data point X_t , and the hidden state of the previous step h_{t-1} . The cell consists of three gates: forget, input, and output, together with an internal cell state C_t , which acts as memory [4]. These gates determine how information is updated and propagated through the sequence.

The forget gate decides which past information to retain, while the input gate determines what new information to store. The cell state is then updated as a combination of these contributions. Finally, the output gate regulates the information passed to the next hidden state, h_t . The gating operations are mathematically defined as follows:

$$f_t = \sigma(W_f \cdot [h_{t-1}, X_t] + b_f), \quad (1)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, X_t] + b_i), \quad (2)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, X_t] + b_C), \quad (3)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \quad (4)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, X_t] + b_o), \quad h_t = o_t \odot \tanh(C_t), \quad (5)$$

where σ denotes sigmoid activation, $\tanh$ is hyperbolic tangent activation, $\odot$ represents element-wise multiplication, X_t is the input vector at time t , h_{t-1} is the previous hidden state, and C_t is the updated cell state. Through this structure, LSTMs can effectively capture both short-term and long-term dependencies, making them highly applicable to modeling the complex temporal dynamics of stock markets [16].

B. TRANSFORMER AND SENTIMENT ANALYSIS

Transformer architectures, introduced by Vaswani et al. in their seminal work "Attention Is All You Need" [11], have become a cornerstone of modern sequence modeling. Unlike recurrent networks, which process sequences step-by-step, Transformers employ an attention-based mechanism that enables parallelization and efficient capture of long-range dependencies. Recent applications in finance have shown their suitability for predicting stock trends [12].

1) TRANSFORMER ARCHITECTURE

The key innovation of the Transformer is the *multi-head self-attention* mechanism [11]. At each layer, the model learns to assign weights to different parts of the input sequence, allowing it to focus selectively on the most relevant tokens. For textual data, such as financial news headlines, this enables the identification of words and phrases that carry critical sentiment.

2) SENTIMENT ANALYSIS IN FINANCIAL FORECASTING

Although historical stock prices reflect market behavior, they do not directly capture the impact of qualitative factors such as news sentiment, public perception, or investor psychology. Sentiment analysis provides a structured means of quantifying these qualitative drivers. Using pre-trained financial language models such as FinBERT tone [6], textual news data can be transformed into sentiment scores that reflect positive, neutral, or negative market sentiment. These scores are particularly important in short-term forecasting, where sudden changes in sentiment often trigger sharp price fluctuations [7].

3) HYBRID INTEGRATION WITH LSTM

The fusion of Transformer-based attention and sentiment analysis with LSTM networks provides a holistic modeling approach. LSTMs are adept at capturing temporal dependencies in price movements [4], [5], while Transformers highlight the most influential textual cues in parallel [11], [12], and sentiment scores quantify market mood through financial text analysis [6], [7]. When combined, these components complement each other: the LSTM processes sequential dependencies in stock prices, the Transformer assigns attention weights to critical textual signals, and sentiment analysis adjusts the overall predictive weighting of features.

This hybrid approach allows the model to capture subtle interactions that would otherwise be missed. For example,

while historical data might suggest an upward trend for a stock, a sudden surge in negative news sentiment—captured by the transformer and sentiment analysis modules—may signal a potential reversal. By explicitly modeling both numerical and qualitative data, the proposed framework offers a more comprehensive and robust predictive capability for financial forecasting [16], [17].

In summary, the integration of Transformer attention and sentiment analysis equips the model with a dual perspective: the quantitative strength of sequential price modeling and the qualitative depth of market sentiment. This synergy improves the accuracy of the forecast, particularly for short-term stock prediction tasks, where market behavior is often influenced as much by perception and news as by historical trends [6], [12]. Hybrid models remain an active research direction; for example, Jiang et al. [18] introduced an attention-based CNN-LSTM with XGBoost, highlighting the value of combining multiple architectures to improve predictive performance.

C. FEDERATED LEARNING

Federated learning (FL), first introduced in [8] and later expanded in [9], has emerged as a promising paradigm for collaborative machine learning, particularly in domains where data privacy and security are critical. Unlike conventional centralized learning approaches that require aggregating data into a single repository, FL enables multiple clients (e.g., devices, institutions, or organizations) to train models locally on their data while only sharing model parameters or gradients with a central server. The global model is then updated by aggregating these local contributions, thereby preserving the confidentiality of the raw data, as illustrated in Fig. 2.

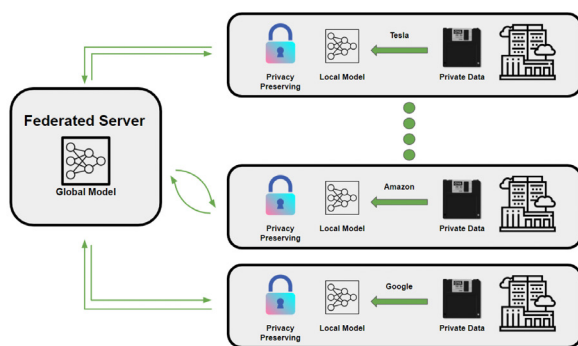


FIGURE 2. Federated learning framework for secure and collaborative stock market prediction across 10 financial institutions, adapted from [8], [9]. Local models are trained on private data, while the federated server aggregates updates to construct a global model.

In financial forecasting, privacy is an essential concern. Market institutions and investors often handle highly sensitive data, and direct exchange of such information for centralized training is rarely feasible due to confidentiality, regulatory, and security restrictions. Federated learning

provides a compelling solution by allowing multiple financial entities to collaboratively train robust models without exposing proprietary or client-specific data [9]. This decentralized approach takes advantage of the collective diversity of data sets between different institutions, improving generalizability while maintaining strict privacy safeguards.

Formally, the global model parameters w are iteratively updated by weighted aggregation of local updates w_i from participating clients, expressed as:

$$w_{t+1} = \sum_{i=1}^K \frac{n_i}{n} w_i^t, \quad (6)$$

where K is the number of clients, n_i represents the number of data samples on the client i , and $n = \sum_{i=1}^K n_i$ is the total number of samples in all clients. This formulation ensures that clients with larger datasets contribute proportionally more to the global model update [8].

In this work, FL was simulated using data sets from ten major companies (Tesla, Apple, Airbnb, Amazon, FedEx, IBM, Microsoft, Nvidia, Oracle, and Google). Each company's stock history was treated as an independent client dataset, from which local LSTM models were trained. The simulated FL framework enhanced robustness by combining heterogeneous data sources, improved generalization between diverse market behaviors, and preserved privacy through noncentralized data handling. Although the current implementation is a simulation, it serves as a proof of concept for deploying privacy-aware forecasting frameworks in multiinstitutional financial environments. The model parameters were then aggregated into a global model following the FedAvg protocol [8]. This setup mimics a real-world deployment in which financial institutions collaboratively improve forecasts while maintaining confidentiality of their internal records [9].

Federated learning was implemented as a simulated environment in which each company's historical stock dataset was treated as an independent client. Specifically, data from the ten selected companies (Tesla, Apple, Airbnb, Amazon, FedEx, IBM, Microsoft, Nvidia, Oracle, and Google) were not pooled centrally but instead used to train local models on each client. Each client trained a local instance of the LSTM-based model using its own data and shared only model parameters with a central server.

Model aggregation was performed using the Federated Averaging (FedAvg) algorithm, where local model weights were averaged proportionally to the size of each client's dataset to update the global model. This aggregation was conducted after each communication round. In our implementation, one communication round corresponds to a full local training cycle on each client followed by global aggregation. In our experiments, convergence was typically achieved after a moderate number of communication rounds. The federated learning setup enables collaborative model training while preserving data privacy, as raw financial data remain local to each client throughout the training process.

III. PROPOSED METHOD

The proposed framework is designed as a modular hybrid architecture in which each component addresses a specific limitation of stock price forecasting. The LSTM module captures nonlinear temporal dependencies in historical price sequences. The Transformer-based multi-head attention layer complements the LSTM by strengthening long-range dependency modeling and enabling the network to emphasize informative time steps. Sentiment features extracted using FinBERT provide exogenous contextual information from financial news that cannot be inferred from price data alone. Federated learning governs the training process by enabling decentralized optimization across heterogeneous client datasets, improving robustness and preserving data privacy. Together, these components form an integrated pipeline rather than independent modules, with each contributing uniquely to predictive performance.

A. DATASET DESCRIPTION AND ANALYSIS

To investigate the predictive performance of different models, we collected historical financial data from Yahoo Finance [19] for ten publicly traded companies: Tesla, Apple, Airbnb, Amazon, FedEx, IBM, Microsoft, Nvidia, Oracle, and Google. These firms were selected due to their importance in the technology and logistics sectors, their high trading volumes, and their diverse volatility profiles, which together provide a robust basis for the evaluation of forecast models [17].

Two different time horizons were analyzed. For long-term forecasts, we retrieved the complete trading history of each company up to January 4, 2024. This allowed us to evaluate the ability of classical and deep learning methods to generalize over extended horizons (Fig. 3). For mid- and short-term forecasts, we restricted the dataset to a six-month window spanning July 6, 2023, to January 4, 2024 (127 trading days), as shown in Fig. 4, allowing us to test the hypothesis with shorter data windows that yield superior accuracy for volatile financial series [10], [20].

	Date	Open	High	Low	Close	Adj Close	Volume
0	2010-06-30	1.719333	2.028000	1.553333	1.588667	1.588667	257806500
1	2010-07-01	1.666667	1.728000	1.351333	1.464000	1.464000	123282000
2	2010-07-02	1.533333	1.540000	1.247333	1.280000	1.280000	77097000
3	2010-07-06	1.333333	1.333333	1.055333	1.074000	1.074000	103003500
4	2010-07-07	1.093333	1.108667	0.998667	1.053333	1.053333	103825500
...	...	...	...	...	...	...	...
3397	2023-12-28	263.660004	265.130005	252.710007	253.179993	253.179993	113619900
3398	2023-12-29	255.100006	255.190002	247.429993	248.479996	248.479996	100615300
3399	2024-01-02	250.080002	251.250000	244.410004	248.419998	248.419998	104654200
3400	2024-01-03	244.979996	245.679993	236.320007	238.449997	238.449997	121082600
3401	2024-01-04	239.250000	242.699997	237.729996	237.929993	237.929993	102629300

FIGURE 3. Long-term historical data for Tesla, spanning June 30, 2010 to January 4, 2024.

The data sets contain six quantitative features: *Open*, *High*, *Low*, *Close*, *Adjusted Close*, and *Volume*, with *Date* serving as

	Date	Open	High	Low	Close	Adj Close	Volume
3275	2023-07-06	278.089996	279.970001	272.880005	276.540009	276.540009	120332100
3276	2023-07-07	278.429993	280.779999	273.769989	274.429993	274.429993	113602000
3277	2023-07-10	276.470001	277.519989	265.100006	269.609985	269.609985	119425400
3278	2023-07-11	268.649994	270.899994	266.369995	269.790009	269.790009	91972400
3279	2023-07-12	276.329987	276.519989	271.459991	271.989990	271.989990	95672100
...	...	...	...	...	...	...	...
3397	2023-12-28	263.660004	265.130005	252.710007	253.179993	253.179993	113619900
3398	2023-12-29	255.100006	255.190002	247.429993	248.479996	248.479996	100615300
3399	2024-01-02	250.080002	251.250000	244.410004	248.419998	248.419998	104654200
3400	2024-01-03	244.979996	245.679993	236.320007	238.449997	238.449997	121082600
3401	2024-01-04	239.250000	242.699997	237.729996	237.929993	237.929993	102629300

FIGURE 4. Six-month dataset for Tesla (July 6, 2023 – January 4, 2024), used for mid- and short-term forecasts.

the temporal index [21]. Long-term forecasting experiments projected prices up to 30 days ahead, while short-term experiments focused on 5-day forecasts. The results confirmed that using smaller and more recent historical windows improved predictive performance [20], underscoring the importance of data consistency in stock prediction tasks. In particular, our baseline LSTM achieved a trend accuracy of 65.36% in 5-day forecasts using three months of data. For the hybrid model (LSTM enhanced with Transformer attention, sentiment analysis, and federated learning), the window was limited to one month because the NewsAPI provided only 30 days of headlines. Despite this constraint, the hybrid model achieved an average R^2 of 0.91, with an MAE of 0.0106 and an MSE of 0.0844, highlighting its robustness.

1) DATA PREPROCESSING

All datasets were preprocessed prior to model training to ensure numerical stability and reproducibility. Missing values were removed, and all numerical features were normalized using Min–Max scaling to the range [0, 1]. To transform the time series into a supervised learning problem, a sliding-window approach was applied, where each input sample consisted of the previous T trading days and the prediction target corresponded to the next-day closing price.

2) FEATURE CONSTRUCTION AND SELECTION

The primary numerical inputs consist of historical OHLCV features (*Open*, *High*, *Low*, *Close*, and *Volume*), which capture price dynamics and trading activity. Additional technical indicators, such as moving averages, were included to smooth short-term fluctuations and highlight underlying trends. For sentiment-enhanced experiments, daily sentiment scores extracted from financial news using FinBERT were aligned with trading dates and concatenated with numerical features. No manual feature elimination was applied, as deep learning models inherently learn feature relevance during training.

3) TRAINING, VALIDATION, AND WORKFLOW

To prevent look-ahead bias, all experiments employed a chronological data split. Baseline models were trained using an 80/20 train–test split, while the hybrid deep learning framework adopted a 70/15/15 split for training, validation, and testing, respectively. The overall workflow consists of data collection, preprocessing and normalization, sequence construction, sentiment alignment, model training (centralized or federated), and performance evaluation using standard forecasting metrics.

B. MOVING AVERAGES AND OTHER INDICATORS

To enrich the feature space and capture longer-term trends, moving averages (MA) were included as auxiliary indicators. MAs smooth short-term fluctuations, helping models identify underlying trends that may otherwise be obscured by volatility [22]. In particular, simple moving averages and exponential moving averages were used to detect reversal patterns and confirm trend direction [23], [24].

In parallel, we benchmarked several traditional and machine learning forecasting models against the LSTM baseline:

- Random Forest Regression: An ensemble of decision trees capable of modeling nonlinear interactions between stock characteristics [2].
- Prophet: A decomposable additive model developed by Meta that captures trend and seasonality, though it struggles with abrupt market shifts [15].
- Support Vector Regression (SVR): Uses kernel-based methods to capture nonlinear dynamics in stock data [3].
- ARIMA: A classical autoregressive model suitable for stationary time series [1].
- SARIMAX: Extends ARIMA to account for seasonality and exogenous signals [1].
- Simple Exponential Smoothing: Produces smooth forecasts with exponentially decaying weights, suitable for stable datasets [25].
- Holt-Winters Exponential Smoothing: Extends exponential smoothing to capture both trend and seasonality [26], [27].

These models served as important baselines. However, their performance was generally weaker than that of LSTM, particularly for short-term forecasting. Our experiments confirmed that LSTM consistently outperformed these methods, achieving higher trend accuracy and lower error metrics. During hyperparameter exploration, we also tested various activation functions, finding LeakyReLU superior to Sigmoid, tanh, and ReLU, due to its ability to mitigate vanishing gradients while modeling continuous values [28]. The key hyperparameters used consistently across all experiments are summarized in Table 1.

C. FOCUSED ANALYSIS ON TESLA DATASET

To perform a more detailed comparison, we selected Tesla (TSLA) as a representative case study. Tesla’s stock is

TABLE 1. Key hyperparameters used in the experiments.

Component	Value
LSTM units	64
Number of LSTM layers	1
Attention type	Multi-head self-attention
Number of attention heads	4
Dropout rate	0.2
Batch normalization	Yes
Activation function	LeakyReLU
Sequence length (T)	20 trading days
Batch size	32
Optimizer	Adam
Learning rate	0.001
Loss function	Mean Squared Error (MSE)
Training epochs	50
Federated aggregation	FedAvg

characterized by high volatility, frequent news coverage, and a long trading history, making it an ideal benchmark to evaluate forecast methods [20], [29]. Using this data set, we were able to highlight the strengths and weaknesses of traditional approaches (e.g., ARIMA, Random Forest, Prophet) relative to LSTM [1], [2], [15].

Although the LSTM model was applied across the ten companies, Tesla provided deeper insights into model behavior under high-volatility conditions. Long-term forecasts revealed that error metrics deteriorated with larger horizons, whereas short-term 5-day predictions yielded higher trend accuracy [20], [21]. This analysis confirmed that LSTM-based methods are particularly effective in capturing the near-term direction of highly dynamic stocks such as Tesla [16], [17].

D. NEWS HEADLINES FOR SENTIMENT ANALYSIS

In addition to quantitative stock features, qualitative market sentiment was integrated into the prediction framework. NewsAPI was used to retrieve financial headlines [30], which were then processed using FinBERT-tone, a pre-trained language model specialized in financial sentiment classification [6]. Each headline was assigned a sentiment score (positive, neutral, or negative), and these scores were aligned with the corresponding trading days [7].

1) SENTIMENT DATA SOURCE AND PROCESSING

NewsAPI provides access to recent, publicly available financial news articles associated with each company. Due to API constraints, sentiment data were limited to the most recent 30 days. Each retrieved headline was processed using the FinBERT-tone model, which is specifically trained on financial text and outputs sentiment probabilities reflecting market tone.

Fig. 5 illustrates the integration of sentiment scores with historical stock features. By combining textual sentiment with numerical indicators, the model captured both market fundamentals and investor psychology [31], [32]. This fusion allowed the model to react to sudden news-driven shocks,

which are often poorly reflected in historical time series alone.

Date	Open	High	Low	Close	Adj Close	Volume	sentiment_positive	sentiment_neutral	sentiment_negative
2023-12-11	-0.764383	-1.092344	-0.895237	-1.094194	-1.094194	0.735566	0.886077	7.556485e-06	0.113915
2023-12-12	-1.271482	-1.638720	-1.339923	-1.450870	-1.450870	-0.870829	0.771475	2.236879e-01	0.004837
2023-12-13	-1.799114	-1.477877	-2.044215	-1.152988	-1.152988	1.794988	0.436152	2.644239e-01	0.299424
2023-12-14	-0.948336	0.189493	-0.480363	0.383459	0.383459	2.555786	0.997672	2.167144e-05	0.002306
2023-12-15	0.260667	0.220188	0.452484	0.703551	0.703551	1.242265	0.930649	5.226901e-02	0.017082
2023-12-18	0.571690	0.786208	0.832578	0.518028	0.518028	0.232379	0.741102	1.232745e-04	0.258774
2023-12-19	0.535383	0.737096	1.037530	1.189570	1.189570	-0.739973	0.992385	4.074478e-04	0.007207
2023-12-20	0.889976	0.921268	0.291006	-0.127384	-0.127384	0.686491	0.594985	2.030768e-01	0.201938
2023-12-21	0.344170	0.302452	0.483577	0.834202	0.834202	0.124523	0.999520	9.486287e-07	0.000479
2023-12-22	0.832335	0.722363	0.833819	0.578126	0.578126	-0.79564	0.756559	1.999968e-01	0.043344
2023-12-26	0.657616	0.691668	1.025110	1.109871	1.109871	-1.312145	0.508208	1.903291e-04	0.491602
2023-12-27	1.124758	1.351003	1.597734	1.740914	1.740914	-0.286686	0.563656	4.357787e-01	0.000565
2023-12-28	1.767380	1.570782	1.000267	0.661742	0.661742	0.086078	0.574486	7.440102e-05	0.425440
2023-12-29	0.731439	0.350336	0.344417	0.047687	0.047687	-0.594245	0.985924	4.229334e-04	0.013653
2024-01-02	0.123912	-0.133422	-0.030708	0.039848	0.039848	-0.382954	0.611635	2.639754e-04	0.388101
2024-01-03	-0.493297	-0.817315	-1.035597	-1.262734	-1.262734	0.476482	0.329115	6.604013e-01	0.010484
2024-01-04	-1.186747	-1.183203	-0.860457	-1.330673	-1.330673	-0.488885	0.698159	3.988885e-03	0.297853
2024-01-05	-1.475388	-1.499978	-1.211982	-1.388157	-1.388157	-1.025098	0.999996	1.200312e-06	0.000003

FIGURE 5. Combined dataset integrating historical stock data with sentiment scores from financial news headlines.

2) INTEGRATION WITH NUMERICAL FEATURES

Daily sentiment scores were temporally aligned with corresponding trading days and concatenated with the normalized OHLCV features and technical indicators to form an extended input vector for the LSTM-based model. This integration enables the model to incorporate exogenous textual information that is not directly observable from historical price data alone.

Due to limitations of the NewsAPI, sentiment scores were restricted to the most recent 30 days of headlines. This constraint confined sentiment-enhanced forecasts to short-term horizons. However, integration of sentiment features improved the responsiveness of the hybrid model to market shocks, complementing the temporal modeling power of LSTM and the contextual focus of Transformer's attention [16], [17]. In future work, large-scale archived news datasets could be integrated to extend sentiment analysis to mid- and long-term horizons.

The proposed methodological choices are guided by the characteristics of financial time series, which exhibit non-linearity, volatility, and sensitivity to external information. LSTM networks address temporal dependencies in price data, while attention mechanisms enhance the modeling of long-range relationships. Sentiment features provide complementary exogenous context, and federated learning enables collaborative training under privacy constraints. Together, these design decisions form a coherent framework that balances predictive performance, robustness, and practical deployment considerations.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. EVALUATION METRICS

To rigorously assess the performance of our forecasting models, we employ four complementary statistical metrics. Each provides a different perspective on the accuracy and robustness of the model, ensuring that our evaluation

captures both the numerical precision and the directional reliability [17], [24].

The first metric, the determination coefficient (R^2), measures the proportion of variance in the target variable (stock price) that is explained by the model. An R^2 value of 1 indicates a perfect fit, while a value close to 0 suggests that the model does not explain the observed variability. In financial forecasting, a high R^2 implies that the model effectively captures the underlying market patterns [20], [21]. It is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (7)$$

The Mean Squared Error (MSE) quantifies the average squared deviation between predicted and actual values:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2. \quad (8)$$

MSE penalizes larger errors more heavily, which makes it particularly relevant in financial contexts where occasional sharp deviations (e.g., during market shocks) can be costly [1], [24].

The Mean Absolute Error (MAE) complements the MSE by capturing the average magnitude of errors without squaring them:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (9)$$

MAE provides an intuitive measure of model precision, reflecting the average dollar deviation between predicted and actual closing prices [17].

Finally, to evaluate the accuracy of the direction, we adopted the mean prediction accuracy metric (MPA) introduced in DP-LSTM [10]. Unlike R^2 , MSE and MAE, which emphasize numerical closeness, MPA assesses whether the model correctly predicts the upward or downward trend of stock prices over a forecast horizon. This is critical for financial decision making, where capturing the correct direction often matters more than minimizing small numerical errors. MPA is defined as

$$\text{MPA}_t = 1 - \frac{1}{L} \sum_{l=1}^L \left| \frac{X_{t,l} - \hat{X}_{t,l}}{X_{t,l}} \right|, \quad (10)$$

where $X_{t,l}$ and $\hat{X}_{t,l}$ represent the actual and predicted stock prices at time t for the horizon l , and L is the length of the forecast window.

In financial forecasting, correctly predicting the direction of price movement is often more critical than minimizing small numerical errors. Directional accuracy directly reflects the model's potential usefulness for trading decisions, portfolio allocation, and risk management, where incorrect trend predictions can lead to disproportionate financial losses. By jointly reporting numerical error metrics (MAE, MSE, and R^2) and trend prediction accuracy, we provide a more

comprehensive assessment of both predictive precision and practical market relevance.

Most baseline models (e.g., ARIMA, Random Forest, SVR) were trained on 80% of the dataset and tested on the remaining 20%, using a chronological division where earlier observations were used for training and the most recent data were reserved for testing [17]. This prevents look-ahead bias and better reflects real forecasting conditions. Prophet, due to its design as a time series decomposition model, does not employ a traditional train–test split, but we aligned its forecast horizon with the other models for comparability [15].

For our hybrid framework that integrates LSTM, Transformer attention, sentiment analysis, and federated learning, we adopted a chronological split of 70/15/15 for training, validation, and testing, respectively. The validation window was drawn from intermediate time periods to enable hyperparameter tuning without exposing the model to future test data. The inclusion of a separate validation set was critical in mitigating overfitting and ensuring that the results generalize effectively to unseen market conditions [33]. All models were evaluated with identical forecast horizons (30 days and 5 days), ensuring fairness and consistency in cross-model comparisons.

To mitigate overfitting, several regularization strategies were employed. Dropout layers were applied within the network architecture to reduce co-adaptation of neurons, and batch normalization was used to stabilize training. A chronological train–validation–test split was adopted, with validation loss monitored during training to assess generalization performance. Training was stopped once convergence was observed and no further improvement in validation loss was achieved, reducing the risk of overfitting.

B. PROPHET

Prophet, a forecasting model developed by Meta (formerly Facebook), is based on an additive decomposition of trend, seasonality, and holidays [15]. Its strengths lie in handling missing data, irregular time series, and strong seasonal patterns, making it a popular choice for business and economic forecasting [34]. However, applying Prophet to financial data presents challenges, as stock prices are often highly volatile, nonlinear, and weakly seasonal [17].

When Prophet was applied to Tesla's full historical dataset, the model produced an overly smooth and linear trend, as shown in Fig. 6. The 30-day forecast demonstrated significant generalization, but did not capture short-term fluctuations. Furthermore, the confidence interval of the forecast (Fig. 7) was excessively wide, ranging from \$197.36 to \$263.62, reducing its utility for practical decision-making. The decomposed seasonality components (Fig. 8) The 30-day forecast demonstrated significant generalization but failed to capture short-term fluctuations [20].

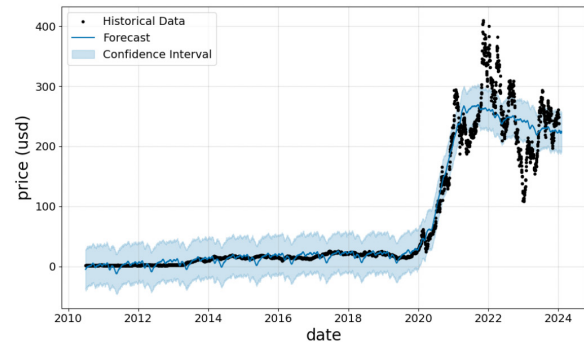


FIGURE 6. Prophet's 30-day forecast using the complete Tesla dataset, displaying overgeneralization.

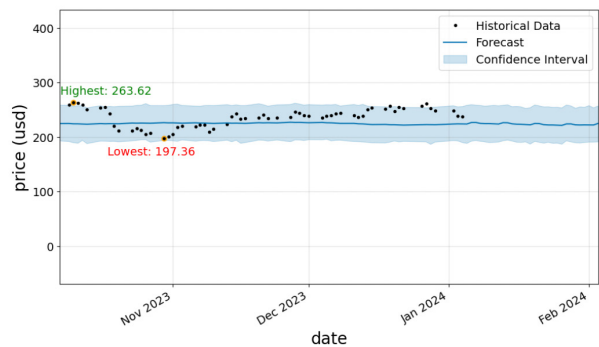


FIGURE 7. Excessive confidence interval of Prophet's 30-day forecast, showing limited predictive precision.

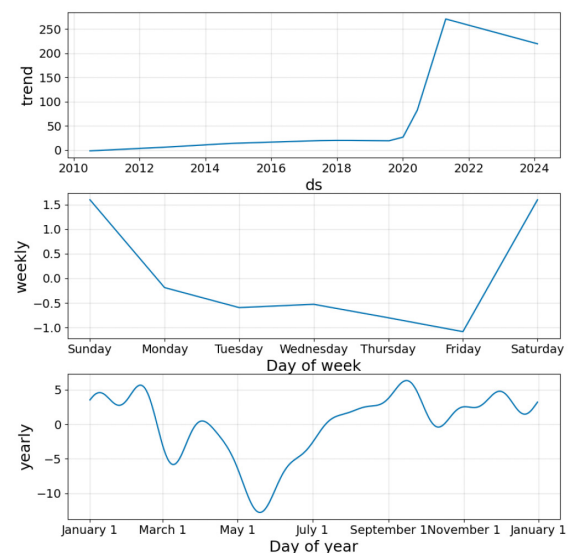


FIGURE 8. Linear seasonality pattern derived from the complete dataset, misaligned with stock market behavior.

To improve predictive precision, we reduced the input window to the most recent six months of Tesla stock prices.

Using 6 months of data narrowed the confidence intervals and produced more realistic nonlinear forecasts (Figs. 9 and 10).

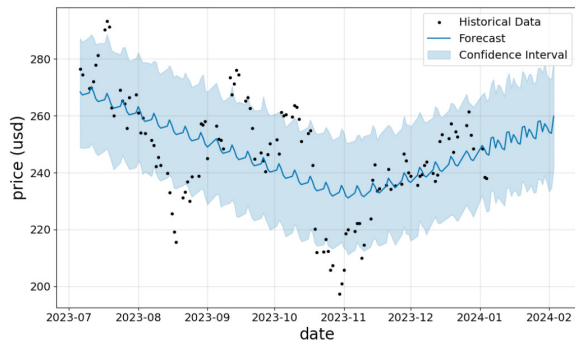


FIGURE 9. Improved 30-day forecast using 6 months of data, showing a more realistic confidence interval and non-linear trend.

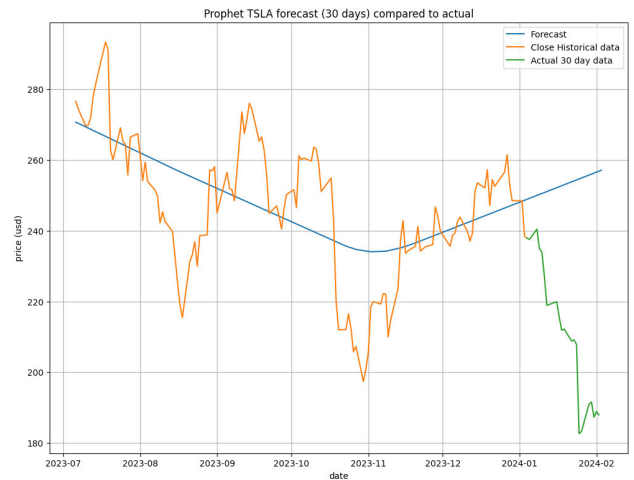


FIGURE 11. Discrepancy between Prophet's 30-day forecast and actual market performance over a 6-month period.

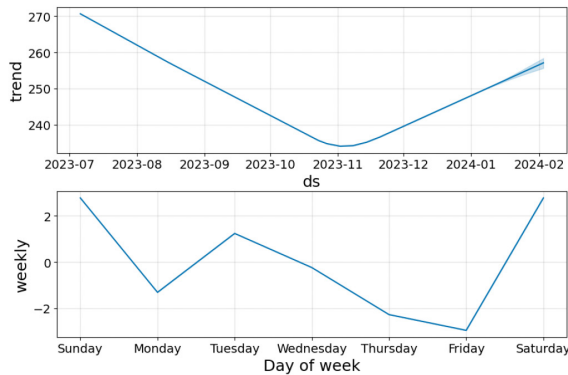


FIGURE 10. Trend and weekly seasonality components from the 6-month data window.

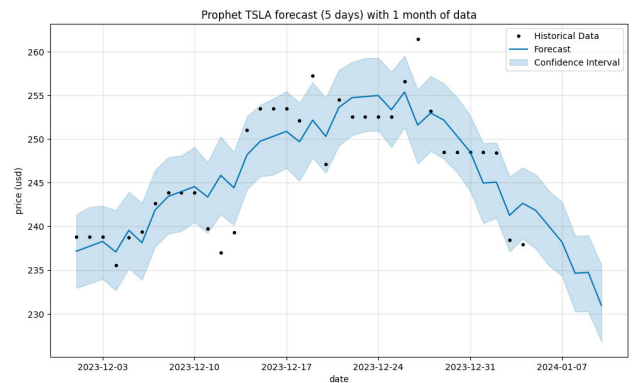


FIGURE 12. 5-day forecast using 1 month of data with adjusted values for weekends.

However, Prophet still underperformed during periods of abrupt market changes, as highlighted in Fig. 11, confirming the model's sensitivity to sudden shocks [21].

Recognizing the importance of recent data for volatile markets, we narrowed the input further, using only one month of historical data to forecast the next five trading days. This short-term setup significantly improved responsiveness, although it introduced a technical issue: Prophet does not account for weekends, resulting in unrealistic negative predictions. By interpolating weekend values, we corrected for this issue and achieved a plausible 5-day forecast, as shown in Fig. 12. The decomposed trend and seasonality components are presented in Fig. 13, while Fig. 14 compares predicted versus actual prices.

The short-term Prophet model produced forecasts closer to the actual trajectory, highlighting the importance of selecting appropriate input windows. However, the model score R^2 was negative ($R^2 = -0.7126$), indicating that its explanation for variance was worse than that of a naïve mean predictor. Although Prophet could capture general directional

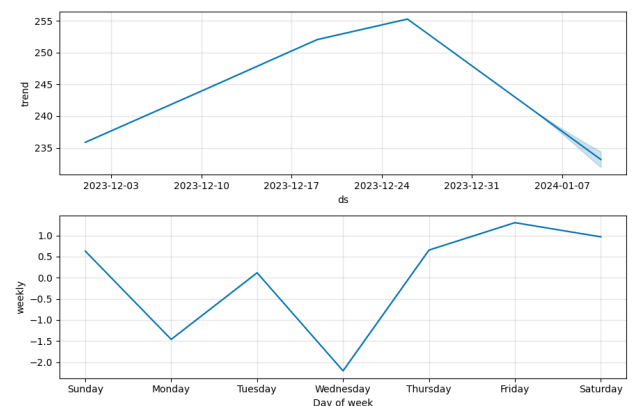


FIGURE 13. Trend and seasonality analysis of Prophet's 5-day forecast using 1 month of data, detailing day-of-week effects.

movement, it consistently failed to model volatility and sharp market turns [24].

In general, the Prophet experiments underscore two key insights: (1) short-term windows are essential for capturing

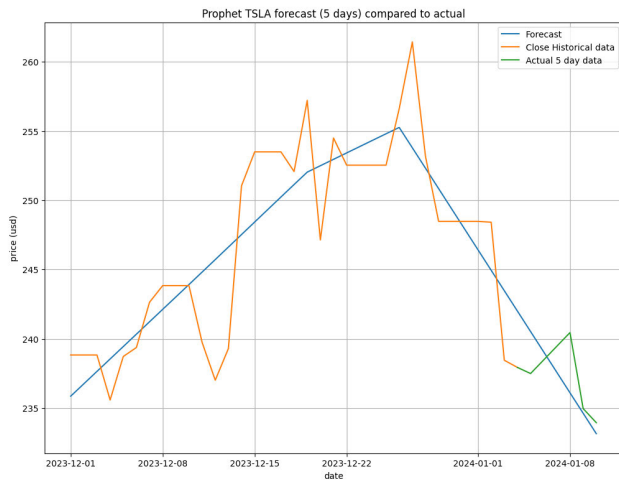


FIGURE 14. Prophet's 5-day forecast accuracy compared to actual stock prices, highlighting the model's predictive capabilities over a short term.

stock dynamics, and (2) the Prophet's additive structure struggles with the noise and unpredictability of financial time series [17]. These limitations motivated our transition toward deep learning methods, where models such as LSTM are better equipped to learn nonlinear temporal dependencies. In the following subsections, we compare Prophet with other machine learning techniques, including Random Forest Regression and Support Vector Regression, to further contextualize the advantages of LSTM.

C. OTHER MACHINE LEARNING TECHNIQUES FOR STOCK PRICE PREDICTION

To benchmark our deep learning approach, we evaluated several established forecasting techniques widely used in financial time series analysis. Random Forest, Support Vector Regression (SVR), SARIMAX, ARIMA, Simple Exponential Smoothing (SES), and Holt-Winters Exponential Smoothing. Each model was tested on Tesla stock for a 30-day forecast, and the results are illustrated in Figs. 15–20.

Random Forest, a group learning method, aggregates multiple decision trees to capture nonlinear dependencies among features [2]. As shown in Fig. 15, it provided stable forecasts and captured the overall trend direction of Tesla's stock. However, its predictions were overly smooth, limiting its ability to capture sharp price fluctuations, a critical limitation in highly volatile markets [17].

Support Vector Regression (SVR) seeks optimal hyperplanes in high-dimensional feature space, making it suitable for nonlinear regression tasks [3]. Its application to Tesla data (Fig. 16) demonstrated that SVR can capture short-term variations, but is highly sensitive to the choice of kernel and regularization parameters. Without careful tuning, the model produced unstable results, highlighting its fragility in practical stock forecasting [17].

SARIMAX extends ARIMA by incorporating both seasonality and exogenous variables [1]. In Fig. 17, SARIMAX

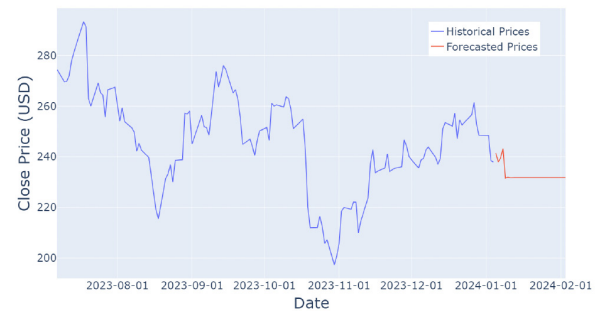


FIGURE 15. Random Forest, Tesla stock 30-day forecast.

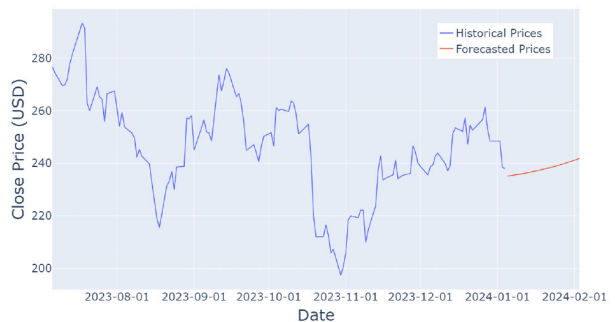


FIGURE 16. Support Vector Regression (SVR), Tesla stock 30-day forecast.

produced consistent cyclical patterns, reflecting its ability to model seasonality. However, its linear assumptions restricted its adaptability to sudden non-linear shifts in Tesla's price.

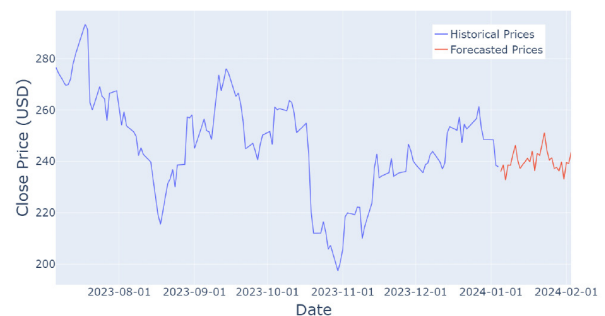


FIGURE 17. SARIMAX, Tesla stock 30-day forecast.

ARIMA, a cornerstone of time series analysis, was also tested as a baseline [1]. Fig. 18 shows its forecasts for Tesla, which were overly simplified and failed to account for seasonal or trend-driven variations. The reliance on linear stationarity assumptions made ARIMA ill-suited to simulate Tesla's highly volatile behavior [24].

Simple Exponential Smoothing (SES) generates forecasts by applying exponentially decaying weights to past observations [25]. As shown in Fig. 19, SES effectively smoothed the series but did not capture trends or seasonality, thus limiting its applicability to dynamic stock data.

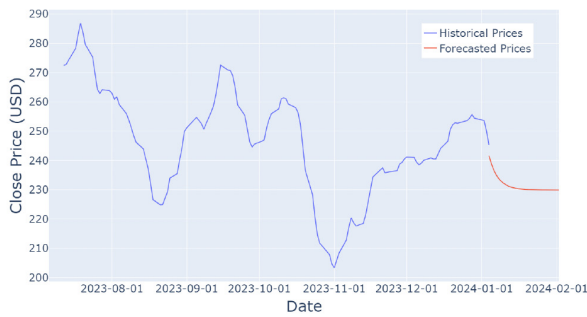


FIGURE 18. ARIMA, Tesla stock 30-day forecast.

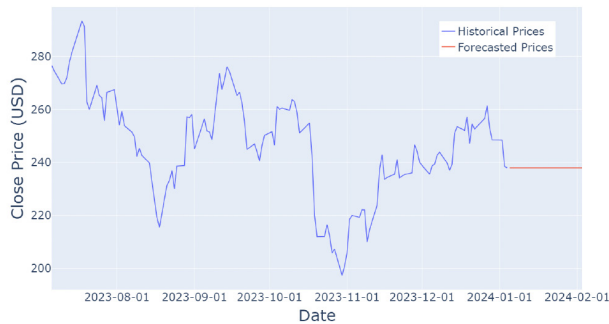


FIGURE 19. Simple Exponential Smoothing (SES), Tesla stock 30-day forecast.

Holt-Winters Exponential Smoothing, which extends SES to incorporate trend and seasonality, performed better, capturing Tesla's seasonal patterns (Fig. 20) [26], [27]. However, its reliance on historical seasonal trends made it slow to adapt to abrupt market shifts, a common feature in financial time series.

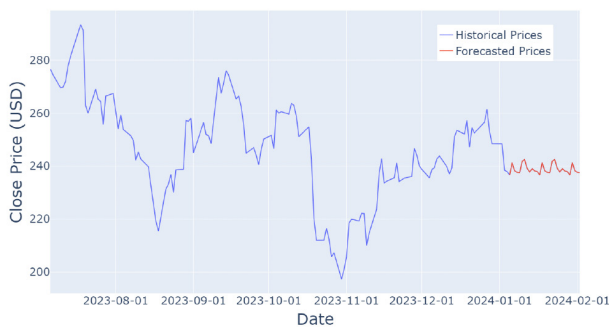


FIGURE 20. Holt-Winters exponential smoothing, Tesla stock 30-day forecast.

Among these methods, Random Forest and SARIMAX emerged as relatively stronger performers, offering reasonable directional forecasts. However, both required extensive parameter tuning or produced repetitive patterns, reducing their suitability for real-time prediction [17]. In general, these traditional approaches demonstrated the limitations of rule-based and linear models when applied to volatile and nonlinear markets such as Tesla. Their shortcomings

highlight the need for more adaptive methods like LSTM networks, which can learn complex temporal dependencies directly from sequential data [20], [21]. In the following subsections, we examine LSTM and its hybrid extensions in greater detail.

As shown in Figs. 26-31 the prediction curves and training-validation loss plots provide insight into how the model tracks market dynamics over time and generalizes to unseen data. In particular, the close alignment between predicted and actual prices in short-term forecasts illustrates the model's ability to capture dominant temporal patterns. The attention mechanism further enhances interpretability by assigning higher importance to informative time steps within the input sequence, enabling the model to focus on relevant historical contexts when generating predictions. Together, these visualizations offer an intuitive understanding of model behavior without introducing additional interpretability overhead.

D. LSTM-BASED APPROACH: ADDRESSING DYNAMIC MARKET BEHAVIOR

Building upon the insights from previous analyzes, particularly the limitations of SARIMAX and Random Forest in adapting to dynamic market changes, we explored LSTM networks. LSTMs are designed to capture temporal dependencies in sequential data and are well-suited to model the complex nonlinear patterns inherent in financial markets [4], [5], [17].

1) MODEL IMPLEMENTATION

The LSTM model was implemented using Python libraries, including NumPy (for numerical computations), pandas (for data preprocessing), and TensorFlow (for deep learning). The stock data was pre-processed in the appropriate supervised learning format before being fed into the model [21]. We used TensorFlow's Sequential API to construct the model, which consisted of the following components:

- **An LSTM layer with 64 units**, captured long-term dependencies in sequences of historical stock data, with sufficient complexity to model nonlinear patterns.
- **Batch Normalization**. Applied after the LSTM layer to stabilize activations, accelerate convergence, and reduce overfitting.
- **A dense output layer with one neuron**. Using a linear activation function, this layer predicted the stock's closing price.

To establish a baseline, we trained and tested the model on Tesla stock using a six-month historical window to predict the following 30 days, consistent with previous work applying LSTMs to financial forecasting [20].

As shown in Fig. 21, the LSTM captured the general direction of Tesla's price movement, but its accuracy diminished over the longer 30-day horizon. This decline reflects two factors: (1) higher volatility in longer time frames and (2) the reduced relevance of older data when forecasting

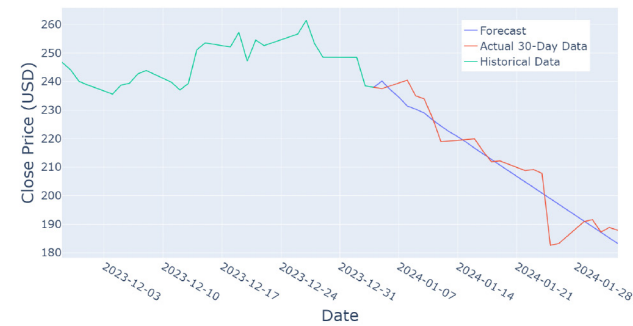


FIGURE 21. Tesla 30-day forecast using LSTM.

into the future. Such challenges illustrate the limitations of applying LSTM directly to extended horizons [16].

2) SHORT-TERM FORECASTING

Encouraged by these insights, we reduced the forecast window to 5 days. As illustrated in Fig. 22, the precision of the model improved markedly in this shorter horizon, achieving a tighter alignment with the actual prices. This reinforces the principle that recent data carry more predictive power in volatile markets and that shorter-term horizons are more suitable for LSTM-based forecasting [21].



FIGURE 22. Tesla 5-day forecast using LSTM.

3) TREND ACCURACY EVALUATION

To further assess performance, we measured the directional accuracy, i.e., whether the model correctly predicted upward or downward trends across all forecasts. We trained the model for 100 runs on each of the ten selected companies and then averaged the results. The LSTM achieved an average trend accuracy of 65.36% (Table 2). This shows that while precise level price predictions remain difficult, the LSTM effectively captures general directionality of the stock market, a metric particularly relevant for trading strategies and risk management [10], [20].

Figure 23 further illustrates the trend accuracy distribution per company. Although the average trend accuracy is moderate, substantial variation exists between different stocks: for instance, Airbnb (94%) and Google (86%) achieved the highest accuracies, while Tesla (34.9%) and IBM (18%)

were considerably more difficult to predict. This highlights that stock-specific volatility and market dynamics strongly influence forecast performance.

TABLE 2. Trend accuracy of LSTM across companies.

Company	Trend Accuracy (%)
Tesla (TSLA)	34.9
Google (GOOGL)	86.0
Apple (AAPL)	71.7
Airbnb (ABNB)	94.0
Amazon (AMZN)	76.0
FedEx (FDX)	41.0
Microsoft (MSFT)	71.0
IBM	18.0
Nvidia (NVDA)	75.0
Oracle (ORCL)	86.0
Average Accuracy	65.36

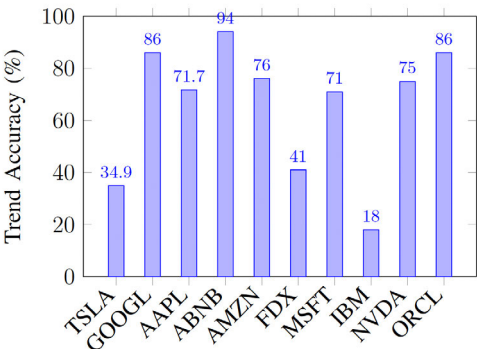


FIGURE 23. LSTM trend accuracy across 10 companies. Average accuracy = 65.36%.

These findings highlight both the potential and limitations of LSTM in financial forecasting. LSTMs are effective in short-term prediction and trend capture, but their accuracy decreases with longer horizons due to market volatility. To address these limitations, we extend our work by integrating Transformer-based attention for improved feature extraction [11], [12], sentiment analysis to incorporate qualitative market drivers [6], and federated learning to preserve data privacy while leveraging distributed datasets [8], [9]. These enhancements are explored in the following sections.

A consolidated comparison of baseline models and the proposed hybrid framework is provided in Table 3, highlighting the relative performance gains of the proposed approach.

E. ENHANCING LSTM WITH ADVANCED TECHNIQUES

Having established the baseline performance of the LSTM, we investigated several advanced techniques aimed at improving accuracy and robustness. Transformer multi-head attention, sentiment analysis, and federated learning.

1) TRANSFORMER MULTI-HEAD ATTENTION

The Transformer multi-head attention mechanism [11], [35] complements the temporal modeling capability of LSTM by

TABLE 3. Performance comparison of baseline models and the proposed hybrid model for short-term(5-day) forecasting. Quantitative metrics are reported where available.

Model	MAE	MSE	R^2	Trend Acc. (%)	Observations
ARIMA	–	–	Low / unstable	–	Linear assumptions; fails under high volatility
SARIMAX	–	–	Moderate	–	Captures seasonality but weak nonlinear adaptation
Prophet	–	–	–0.71	–	Over-smoothing and wide confidence intervals
Random Forest	–	–	Moderate	–	Stable but overly smooth predictions
SVR	–	–	Variable	–	Sensitive to kernel and hyperparameters
SES / Holt–Winters	–	–	Low	–	Unable to capture abrupt market changes
LSTM (baseline)	0.184	0.041	0.865	65.36	Captures short-term temporal dependencies
Proposed Hybrid Model	0.129	0.022	0.933	Improved	Attention-, sentiment-, and FL-enhanced

enabling the network to capture long-range dependencies across sequences. This allows the model to assign varying importance to different time steps, thereby enhancing its ability to identify complex relationships in financial data.

2) SENTIMENT ANALYSIS INTEGRATION

To incorporate qualitative factors, we integrate sentiment features derived from financial news headlines. Using the FinBERT tone model [6], [36], each headline was converted into sentiment scores (positive, neutral, or negative). These features were then combined with stock price data to provide context about investor psychology, which often influences short-term market movements. Due to restrictions of the NewsAPI, sentiment analysis was limited to the most recent 30 days, confining experiments the short-term forecasts.

3) FEDERATED LEARNING SIMULATION

We implemented a simulated FL environment to evaluate the effectiveness of decentralized training for financial forecasting. Each of the ten selected companies (Tesla, Apple, Airbnb, Amazon, FedEx, IBM, Microsoft, Nvidia, Oracle, and Google) was treated as an independent client dataset. Local LSTM models were trained on each dataset, and their weights were aggregated using the FedAvg protocol [8].

This simulation demonstrated three key benefits: (1) robustness, by leveraging heterogeneous client datasets; (2) improved generalization across diverse market behaviors; and (3) privacy preservation, since raw data remained local to each client. Figure 24 presents the loss curves for the FL training and validation epochs, showing stable convergence and improved resistance to overfitting compared to centralized training.

In general, FL simulation confirmed that the integration of decentralized learning into stock prediction can significantly enhance model reliability and adaptability while simultaneously addressing confidentiality concerns in financial markets.

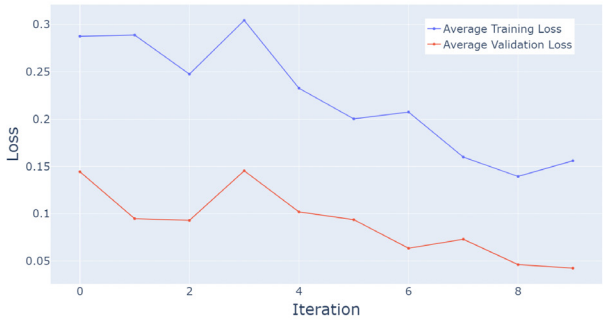


FIGURE 24. Training and validation losses under federated learning simulation.

4) ENHANCED MODEL ARCHITECTURE

The combined model architecture is shown in Fig. 25. It begins with an input layer that processes time series sequences, followed by an LSTM layer to capture temporal dependencies. A dropout layer is applied to prevent overfitting, after which a multi-head attention mechanism highlights the most relevant time steps. The outputs are aggregated using Global Average Pooling, then refined through dense layers, and finally passed to the output layer to generate stock price forecasts. Sentiment features are concatenated with the LSTM-Transformer outputs, and federated learning governs the distributed training process.

5) PERFORMANCE EVALUATION

We first validated the enhanced model on Tesla (TSLA). As shown in Fig. 26, the training and validation losses decreased steadily, indicating good convergence with minimal overfitting. The actual predictions for Tesla (Fig. 27) closely aligned with the ground truth prices, supported by strong evaluation metrics: MSE = 0.039, MAE = 0.179, and $R^2 = 0.888$.

Similar training-validation patterns were observed for other companies, as shown in Figs. 28–30. The final predictions for each stock are illustrated in Figs. 29–31, while

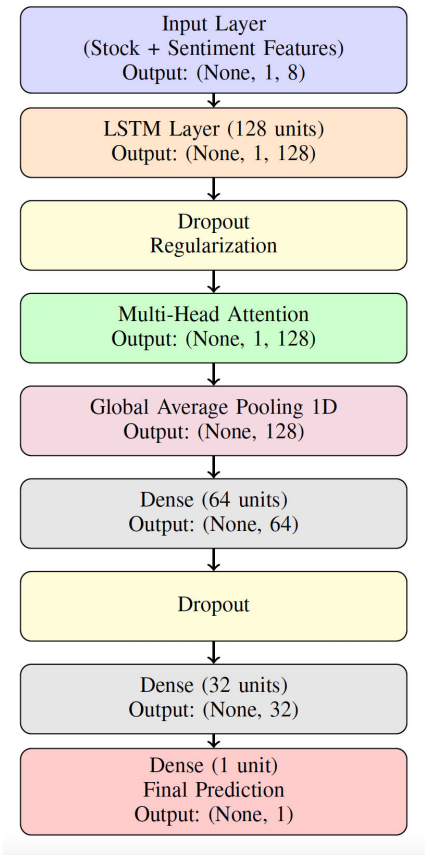


FIGURE 25. Enhanced LSTM architecture incorporating multi-head attention, sentiment features, and federated learning.

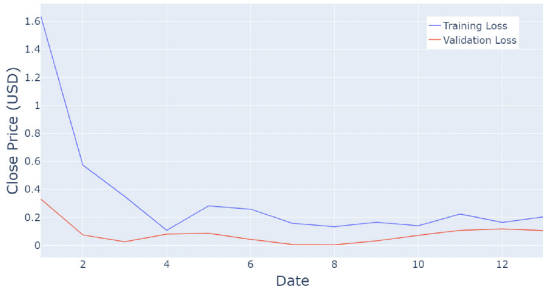


FIGURE 26. Training and validation loss curves for TSLA.

aggregated evaluation metrics are summarized in Table 4. Across all companies, R^2 values exceeded 0.84, with several models exceeding 0.93, underscoring the robustness of the enhanced LSTM framework.

Finally, Fig. 24 shows the loss of federated training versus the loss of validation across epochs. The steady decrease in both confirms the effectiveness of federated learning in stabilizing the training process and improving generalization.

In general, these results demonstrate that enhancing LSTM with attention mechanisms, sentiment features, and federated learning significantly improves stock price prediction. The architecture consistently produced high R^2 values and low

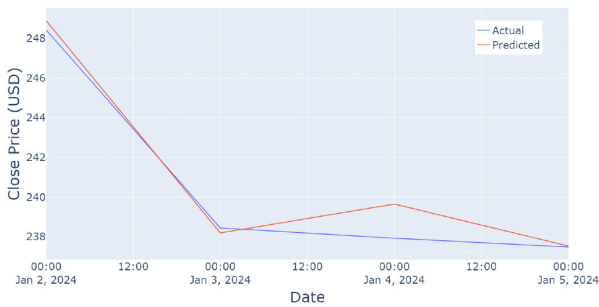


FIGURE 27. Predicted vs. actual TSLA stock prices using the enhanced LSTM model.

TABLE 4. Evaluation metrics for the enhanced LSTM model across different companies.

Company	MSE	MAE	R^2
TSLA	0.039	0.179	0.888
AAPL	0.017	0.114	0.853
ABNB	0.007	0.059	0.845
FDX	0.002	0.038	0.934
NVDA	0.013	0.103	0.968
ORCL	0.005	0.065	0.892
GOOGL	0.017	0.121	0.870
AMZN	0.056	0.207	0.885
MSFT	0.017	0.117	0.938
IBM	0.023	0.111	0.932

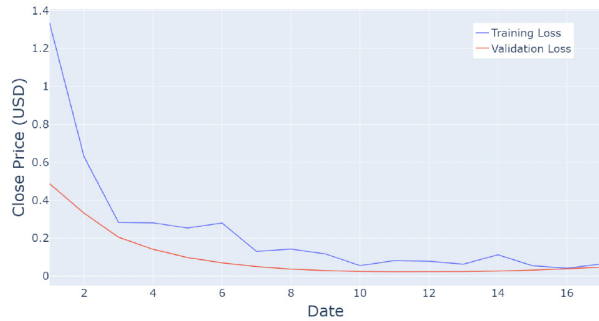


FIGURE 28. Combined LSTM model train - validation evaluation for AAPL data.

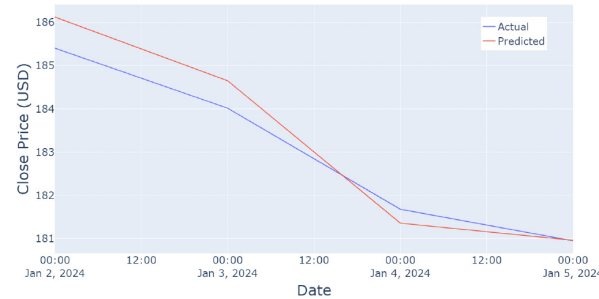


FIGURE 29. Combined LSTM model's actual prediction for AAPL data.

error rates in all companies, demonstrating its promise for robust and privacy-oriented financial forecasting.

The experimental results demonstrate that improvements in predictive accuracy are closely tied to the architectural enhancements introduced in the proposed framework. Short-term forecasting performance benefits from the integration of attention and sentiment features, which improve responsiveness to recent market signals. The observed reduction in error metrics and increase in directional accuracy indicate that the model captures both numerical precision and market-relevant trends, validating the design choices outlined in Section III.

6) ABLATION STUDY

To evaluate the individual contributions of each enhancement, we conducted an ablation study comparing four configurations: (1) baseline LSTM, (2) LSTM with multihead attention, (3) LSTM with attention and sentiment integration, and (4) the full hybrid model incorporating federated learning. The results are summarized in Table 5. To empirically validate the contribution of each architectural component, we conduct an ablation study in which attention mechanisms, sentiment features, and federated learning are progressively added to the baseline LSTM model.

TABLE 5. Ablation study results showing incremental performance gains from model enhancements.

Model Configuration	MSE	MAE	R^2
LSTM (baseline)	0.041	0.184	0.865
LSTM + Attention	0.033	0.161	0.902
LSTM + Attention + Sentiment	0.027	0.143	0.918
Full Hybrid (with Federated Learning)	0.022	0.129	0.933

The study demonstrates clear improvements at each stage. Adding multi-head attention improved the model’s ability to capture long-range dependencies, reducing MSE by nearly 20% compared to the baseline. Incorporating sentiment analysis provided further gains, particularly in short-term predictions, by embedding contextual market signals from news headlines. Finally, simulation of federated learning produced the most robust results, with the lowest error metrics and the highest R^2 , underscoring the value of decentralized training in improving generalization. These incremental improvements confirm that each component of the proposed hybrid model contributes meaningfully to the prediction of performance [37].

The computational cost of the proposed framework primarily arises during the training phase due to the integration of deep sequential models, attention mechanisms, and federated learning. However, inference complexity remains comparable to that of standard LSTM-based models, making the approach feasible for practical short-term forecasting applications. The federated learning component introduces additional communication overhead during training but does not affect inference-time efficiency, as predictions are generated using the aggregated global model. In our experiments, model convergence was achieved within a moderate number of

training epochs, indicating that the added architectural components improve predictive performance without imposing prohibitive computational demands.

F. LIMITATIONS

While the proposed framework demonstrates strong short-term forecasting performance, several limitations should be acknowledged. First, financial markets are inherently volatile and influenced by unpredictable external events, which can reduce model reliability during periods of extreme market stress. Second, sentiment analysis was constrained to a 30-day window due to NewsAPI limitations, restricting sentiment-enhanced predictions to short-term horizons. Third, federated learning was implemented as a simulated environment rather than a real-world multi-institutional deployment, which may limit conclusions regarding communication overhead and scalability. Finally, model performance varied across companies, reflecting differences in volatility and market behavior, which may affect generalization across sectors. These limitations highlight opportunities for future work, including larger-scale sentiment datasets, real-world federated learning deployments, and broader market coverage.



FIGURE 30. Combined LSTM model train - validation evaluation for IBM data.



FIGURE 31. Combined LSTM model's actual prediction for IBM data.

V. CONCLUSION

This study evaluated traditional statistical, machine learning, and deep learning methods for stock market forecasting. LSTM emerged as the strongest baseline, achieving an average trend accuracy of 65.36% and outperforming

Random Forest and SARIMAX, which required extensive tuning. To further enhance predictive power, we integrated Transformer attention, sentiment analysis using FinBERT, and a simulated federated learning framework. The hybrid model consistently achieved high performance, with an average R^2 of 0.91 across innten major companies, some of which exceeded 0.93, confirming the benefits of combining temporal modeling, contextual sentiment and privacy-preserving training.

Future work should expand the analysis to additional sectors, incorporate larger-scale sentiment and macroeconomic datasets, and explore real-world federated learning deployments to further strengthen predictive accuracy and robustness. In addition, integrating explainable AI techniques could improve interpretability and foster trust among financial professionals, making the framework more applicable in decision-support contexts.

From a practical perspective, the proposed framework is particularly suited for short-term forecasting scenarios where timely reactions to market signals are critical, such as portfolio rebalancing, risk monitoring, and decision-support systems for traders and financial institutions. The modular design allows components such as sentiment analysis or federated learning to be enabled or disabled depending on data availability and privacy constraints. Future work may extend this framework by incorporating longer-horizon sentiment datasets, macroeconomic indicators, and real-world federated deployments across financial institutions. Additionally, integrating explainable AI techniques could further enhance transparency and trust, facilitating broader adoption in operational financial environments.

REFERENCES

- [1] G. E. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed., Hoboken, NJ, USA: Wiley, 2015.
- [2] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
- [3] A. J. Smola and B. Schölkopf, "A tutorial on support vector regression," *Statist. Comput.*, vol. 14, no. 3, pp. 199–222, Aug. 2004.
- [4] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.
- [5] K. Greff, R. Srivastava, J. Koutník, B. R. Steunebrink, and J. Schmidhuber, "LSTM: A search space Odyssey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2222–2232, Dec. 2017.
- [6] D. Araci, "FinBERT: Financial sentiment analysis with pre-trained language models," 2019, *arXiv:1908.10063*.
- [7] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artif. Intell. Statist. (AISTATS)*, vol. 54, 2017, pp. 1273–1282.
- [8] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artif. Intell. Statist. (AISTATS)*, 2016, pp. 1273–1282.
- [9] Q. Yang, Y. Liu, T. Chen, and Y. Tong, "Federated machine learning: Concept and applications," *ACM Trans. Intell. Syst. Technol. (TIST)*, pp. 1–19, 2019.
- [10] X. Li, Y. Li, H. Yang, L. Yang, and X. Liu, "DP-LSTM: Differential privacy-inspired LSTM for stock prediction using financial news," *IEEE Access*, pp. 171423–171433, 2019.
- [11] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2025, pp. 5998–6008.
- [12] Y. Zhang, X. Wang, J. Zhang, and Y. Liu, "Transformer-based stock movement prediction with financial news," in *Proc. 30th Int. Joint Conf. Artif. Intell. (IJCAI)*, 2021, pp. 4470–4476.
- [13] W. Yang, C. Zhang, F. Li, and M. Zhao, "An evaluation of deep learning models for stock market prediction," 2024, *arXiv:2408.12408*.
- [14] J. Li, Y. Wang, X. Chen, and L. Huang, "An explainable deep learning approach for stock market trend prediction," *Heliyon*, vol. 10, no. 15, 2024, Art. no. e31439.
- [15] S. J. Taylor and B. Letham, "Forecasting at scale," *Amer. Statistician*, vol. 72, no. 1, pp. 37–45, 2018.
- [16] B. Lim and S. Zohren, "Time-series forecasting with deep learning: A survey," *Phil. Trans. Roy. Soc. A*, vol. 379, no. 2194, 2021, Art. no. 20200209.
- [17] O. B. Sezer, M. U. Gudelek, and A. M. Ozbayoglu, "Financial time series forecasting with deep learning : A systematic literature review: 2005–2019," *Appl. Soft Comput.*, vol. 90, May 2020, Art. no. 106181.
- [18] H. Jiang, Z. Xu, and Q. Liu, "Attention-based CNN-LSTM and XGBoost hybrid model for stock price prediction," 2022, *arXiv:2204.02623*.
- [19] Yahoo Finance. (2024). *Yahoo Finance API*. Accessed: Jan. 2024. [Online]. Available: <https://finance.yahoo.com/>
- [20] T. Fischer and C. Krauss, "Deep learning with long short-term memory networks for financial market predictions," *Eur. J. Oper. Res.*, vol. 270, no. 2, pp. 654–669, Oct. 2018.
- [21] D. M. Q. Nelson, A. C. M. Pereira, and R. A. de Oliveira, "Stock market's price movement prediction with LSTM neural networks," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, May 2017, pp. 1419–1426.
- [22] J. J. Murphy, *Technical Analysis of the Financial Markets: A Comprehensive Guide to Trading Methods and Applications*. New York, NY, USA: New York Institute of Finance, 1999.
- [23] R. G. Brown, *Forecasting, With Emphasis on Applications*. New York, NY, USA: Academic, 2004.
- [24] C. Chatfield, *Time-Series Forecasting*. Boca Raton, FL, USA: CRC Press, 2000.
- [25] R. G. Brown, *Statistical Forecasting for Inventory Control*. New York, NY, USA: McGraw-Hill, 1959.
- [26] C. C. Holt, "Forecasting seasonals and trends by exponentially weighted moving averages," *Office Naval Res.*, Arlington, VA, USA, Research Memorandum 52, 1957.
- [27] P. R. Winters, "Forecasting sales by exponentially weighted moving averages," *Manage. Sci.*, vol. 6, no. 3, pp. 324–342, Apr. 1960.
- [28] A. L. Maas, A. Y. Hannun, and A. Y. Ng, "Rectifier nonlinearities improve neural network acoustic models," in *Proc. 30th Int. Conf. Mach. Learn. (ICML)*, vol. 30, no. 1, 2013, pp. 1–8.
- [29] C. Krauss, X. A. Do, and N. Huck, "Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the S&P 500," *Eur. J. Oper. Res.*, vol. 259, no. 2, pp. 689–702, Jun. 2017.
- [30] NewsAPI. (2024). *NewsAPI: JSON API for Live and Historic News Headlines*. Accessed: Jan. 2024. [Online]. Available: <https://newsapi.org/>
- [31] P. C. Tetlock, "Giving content to investor sentiment: The role of media in the stock market," *J. Finance*, vol. 62, no. 3, pp. 1139–1168, 2007.
- [32] J. Bollen, H. Mao, and X. Zeng, "Twitter mood predicts the stock market," *J. Comput. Sci.*, vol. 2, no. 1, pp. 1–8, Mar. 2011.
- [33] C. Bergmeir, R. J. Hyndman, and B. Koo, "A note on the validity of cross-validation for evaluating autoregressive time series prediction," *Comput. Statist. Data Anal.*, vol. 120, pp. 70–83, Apr. 2018.
- [34] K. Bandara, C. Bergmeir, and S. Smyl, "Forecasting across time series databases using recurrent neural networks on groups of similar series: A clustering approach," *Expert Syst. Appl.*, vol. 140, Feb. 2020, Art. no. 112896.
- [35] J. Wu, Q. Sun, W. Xu, and J. Xu, "Deep learning-based stock price prediction using attention mechanism and multi-task learning," *Inf. Sci.*, vol. 512, pp. 279–292, Mar. 2020.
- [36] Y. Yang, Y. Oba, M. Utiyama, E. Sumita, R. Wang, and D. Kawahara, "Finbert: A pretrained language model for financial communications," in *Proc. Conf. Empirical Methods Natural Lang. Process., Findings*, Jun. 2020, pp. 273–282.
- [37] D. Arpit, S. Jastrzebski, N. Ballas, D. Krueger, and Y. Bengio, "On the importance of ablation studies in deep learning," in *NeurIPS Workshop on Critiquing and Correcting Trends in Machine Learning*, 2021.



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